

Review Feedback — Questions Raised

AI-Native Self-Healing O-RAN Network using a Digital Twin · Points raised in the previous review · Tanmay

The questions and comments raised by the reviewer in the previous review:

Q1. Specialize on ONE problem.

The problem statement was too broad and covered several problems at once; it was hard to follow what exactly the project set out to solve. Narrow it down to a single, clearly-defined problem.

Q2. Which domain / layer are you attacking?

During the presentation: is this a routing problem, a physical-layer problem, or something else? Where exactly in the network does the work sit?

Q3. Make the problem statement understandable.

The presentation itself was hard to follow — the core problem did not come across clearly to the audience.

Purpose of this sheet. Page 2 answers each of these questions and shows how the project was changed to address them — so the improvements made since the last review are on record.

Review Feedback — Questions & How We Addressed Them

What changed since the previous review · Tanmay

Q1. Specialize on ONE problem.

Answer / action taken. We narrowed the whole project to a single problem on the O-RAN fronthaul Synchronization (S-) plane: **tell a benign timing fault from a malicious attack, and heal the clock — verified by a digital twin — before the base station fails (~2 s)**. Everything else (anomaly detection, RCA, general self-healing) is now supporting machinery, not a separate claim. The project has one spine and one sentence anyone can repeat.

Q2. Which domain / layer are you attacking?

Answer. The **Open Fronthaul Synchronization plane (S-plane)** — the timing link between the O-DU and the O-RU, at **Layer 1–2** (PTP / IEEE-1588 over Ethernet, plus SyncE). Stated precisely: **it is NOT the routing/network layer, and NOT the radio air-interface** — it is the fronthaul timing/transport layer.

Attack vector, if pressed: a malicious device on the fronthaul acts as a fake PTP grandmaster (forged Announce/Sync) or replays timing messages, because PTP has no built-in authentication.

Q3. Make the problem statement understandable.

Answer / action taken. We rebuilt the presentation around one clear spine with a plain-language storyline: (1) why fronthaul timing matters (~100 ns budget; a timing attack crashes the base station in ~2 s), (2) prior work detects but does not respond, (3) our problem — discriminate fault vs attack and heal, twin-verified, before failure. Slide titles were made short and proper, and the deck now leads with the worklet, previous reviews, and a pointwise problem statement.

One-line summary for the reviewer. Since the last review we (i) specialized to a single S-plane problem, (ii) named the exact layer (fronthaul S-plane, L1–L2 / PTP + SyncE), and (iii) rebuilt the presentation around one clear, plain-language problem statement.